

Unit 6 Test Study Guide DRAFT

Polynomials | Draft New Format

Name: _____

Period: _____ Due: _____

How to use this guide: This draft keeps the Version A/B structure from last year's Unit 6 guide and updates it into the shared packet format. Each skill has a Version A and Version B so students can try a second similar problem after getting help.

Multiple-Choice Practice

1A. Polynomial Vocabulary | Version A

Which expression is NOT a polynomial? A. $3x^2 - 5x + 1$ B. $7x - 4$ C. $5/x + 2$ D. 9	Work / thinking _____ _____ _____
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1B. Polynomial Vocabulary | Version B

Which expression is NOT a polynomial? A. $x^3 + 2x$ B. $4x^{(1/2)} + 1$ C. $-8x^2$ D. 6	Work / thinking _____ _____ _____
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2A. Classifying Polynomials | Version A

Classify $5x^4 - 3$. A. quadratic binomial B. quartic binomial C. cubic monomial D. quartic trinomial	Work / thinking _____ _____ _____
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2B. Classifying Polynomials | Version B

Classify $-7x^2$. A. quadratic monomial B. quadratic binomial C. linear monomial D. constant	Work / thinking _____ _____ _____
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3A. Greatest Common Factor | Version A

What is the GCF of $12x^3 + 18x^2$?	Work / thinking
A. $3x$	_____
B. $6x^2$	_____
C. $12x^2$	_____
D. $18x^3$	

3B. Greatest Common Factor | Version B

What is the GCF of $15a^4 - 25a^2$?	Work / thinking
A. $5a^2$	_____
B. $10a^2$	_____
C. $15a^4$	_____
D. $25a$	

4A. Standard Form | Version A

Which is $4x - 7 + 3x^2$ in standard form?	Work / thinking
A. $4x + 3x^2 - 7$	_____
B. $3x^2 + 4x - 7$	_____
C. $-7 + 4x + 3x^2$	_____
D. $3x^2 - 7 + 4x$	

4B. Standard Form | Version B

Which is $6 - 2x^3 + 5x$ in standard form?	Work / thinking
A. $-2x^3 + 5x + 6$	_____
B. $5x - 2x^3 + 6$	_____
C. $6 + 5x - 2x^3$	_____
D. $-2x^3 + 6 + 5x$	

5A. Multiplying Binomials | Version A

Expand $(x + 4)(x - 3)$. A. $x^2 + x - 12$ B. $x^2 - x - 12$ C. $x^2 + 7x + 12$ D. $x^2 - 12$	Work / thinking
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5B. Multiplying Binomials | Version B

Expand $(x - 5)(x + 2)$. A. $x^2 - 3x - 10$ B. $x^2 + 7x - 10$ C. $x^2 - 7x + 10$ D. $x^2 - 10$	Work / thinking
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6A. Factoring Trinomials | Version A

Factor $x^2 + 7x + 12$. A. $(x + 3)(x + 4)$ B. $(x - 3)(x - 4)$ C. $(x + 2)(x + 6)$ D. prime	Work / thinking
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6B. Factoring Trinomials | Version B

Factor $x^2 - 9x + 20$. A. $(x - 4)(x - 5)$ B. $(x + 4)(x + 5)$ C. $(x - 2)(x - 10)$ D. prime	Work / thinking
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Free-Response Practice

Show clear work. Use the Version B problem as a second check after you have tried Version A.

7-A. Classifying Polynomials | Version A

Classify $3x^3 - 2x + 8$ by degree and number of terms. Answer: _____	Work / thinking _____ _____ _____ _____ _____
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7-B. Classifying Polynomials | Version B

Classify $-6x^4 + 9x^2$ by degree and number of terms. Answer: _____	Work / thinking _____ _____ _____ _____ _____
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8-A. Adding Polynomials | Version A

Simplify $(4x^2 - 3x + 7) + (2x^2 + 8x - 5)$. Answer: _____	Work / thinking _____ _____ _____ _____ _____
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8-B. Subtracting Polynomials | Version B

Simplify $(7x^2 + 2x - 4) - (3x^2 - 5x + 9)$. Answer: _____	Work / thinking _____ _____ _____ _____ _____
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9-A. Multiplying Polynomials | Version A

Multiply $(2x + 3)(x^2 - 4x + 5)$. Answer: _____	Work / thinking _____ _____ _____ _____ _____ _____
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9-B. Multiplying Polynomials | Version B

Multiply $(x - 2)(3x^2 + x - 6)$. Answer: _____	Work / thinking _____ _____ _____ _____ _____ _____
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10-A. Factoring GCF | Version A

Factor $18x^3 - 24x^2$ completely. Answer: _____	Work / thinking _____ _____ _____ _____ _____
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10-B. Factoring GCF | Version B

Factor $20a^4 + 35a^2$ completely. Answer: _____	Work / thinking _____ _____ _____ _____ _____
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11-A. Factoring Trinomials | Version A

Factor $x^2 + 2x - 24$. Answer: _____	Work / thinking _____ _____ _____ _____ _____
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11-B. Factoring Trinomials | Version B

Factor $x^2 - 5x - 36$. Answer: _____	Work / thinking _____ _____ _____ _____ _____
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12-A. Difference of Squares | Version A

Factor $9x^2 - 49$. Answer: _____	Work / thinking _____ _____ _____ _____ _____
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12-B. Difference of Squares | Version B

Factor $25x^2 - 16$. Answer: _____	Work / thinking _____ _____ _____ _____ _____
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13-A. Perfect-Square Trinomials | Version A

Factor $x^2 + 10x + 25$. Answer: _____	Work / thinking _____ _____ _____ _____ _____
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13-B. Perfect-Square Trinomials | Version B

Factor $x^2 - 14x + 49$. Answer: _____	Work / thinking _____ _____ _____ _____ _____
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14-A. Solving by Factoring | Version A

Solve $x^2 + x - 20 = 0$. Answer: _____	Work / thinking _____ _____ _____ _____ _____
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14-B. Solving by Factoring | Version B

Solve $x^2 - 3x - 28 = 0$. Answer: _____	Work / thinking _____ _____ _____ _____ _____
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Student Self-Check Answer Bank

Important: This is not a worked solution key. It is a self-check bank. If your answer is not here, go back and check the setup, notation, and calculation.

Multiple-Choice Practice

Solve first, then check. Cross off each answer only after your work supports it.

B. quartic binomial	C. $5/x + 2$
B. $6x^2$	A. $-2x^3 + 5x + 6$
A. $x^2 - 3x - 10$	A. $x^2 + x - 12$
B. $3x^2 + 4x - 7$	A. $(x - 4)(x - 5)$
B. $4x^{1/2} + 1$	A. $5a^2$
A. $(x + 3)(x + 4)$	A. quadratic monomial

Free-Response Practice

Solve first, then check. Cross off each answer only after your work supports it.

cubic trinomial	$6x^2(3x - 4)$
$5a^2(4a^2 + 7)$	$x = -4$ or $x = 7$
$(x - 9)(x + 4)$	$3x^3 - 5x^2 - 8x + 12$
$x = -5$ or $x = 4$	$6x^2 + 5x + 2$
$(5x - 4)(5x + 4)$	$(x + 6)(x - 4)$
quartic binomial	$(3x - 7)(3x + 7)$
$(x - 7)^2$	$(x + 5)^2$
$2x^3 - 5x^2 - 2x + 15$	$4x^2 + 7x - 13$

Final Check Before the Assessment

- ☐ I can classify polynomials by degree and number of terms.
- ☐ I can write polynomials in standard form.
- ☐ I can add, subtract, and multiply polynomials.
- ☐ I can factor by GCF, grouping, and trinomial patterns.
- ☐ I can solve polynomial equations by factoring.